

ShanghaiTech University
School of Information Science and Technology

EE115A Lab Experiments
Experiment : Non-Ideal Op-Amps

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1. Introduction

In this Lab 1 & 2, you will characterize an OP07G operational amplifier. In particular, you will be measuring the following non-ideal properties of op-amps: **input offset voltage, slew rate, and finite gain and band width**. Attached are the datasheet and SPICE model for the OP07G op-amp, which you should find helpful while doing the lab and pre-lab. Make sure to download Pre-Lab Worksheet and the Lab Worksheet. Make sure to complete the Pre-Lab Worksheet before coming to lab and send it to [elearning or Email](#) in time. Please notice Pre-Lab 2.1 to 2.2 & 2.6 and Lab 3.1 & 3.2 are for Lab1 and Pre-Lab 2.3 to 2.5 and Lab 3.3 to 3.5 is for Lab2. Fill out the Lab Worksheet while doing the lab and turn it in with supporting traces and plots to [elearning or Email](#) in time as well.

2. Pre-Lab

For the pre-lab, you primarily will be doing Multisim simulations of the measurements that you will be doing in your lab section to help prepare you for the lab. Since OP07GJ circuit model exists in the Multisim library, you can use it directly. Besides, try to read the downloaded spice model or that in Multisim, and the datasheet to learn more about the characteristics of OP07G.

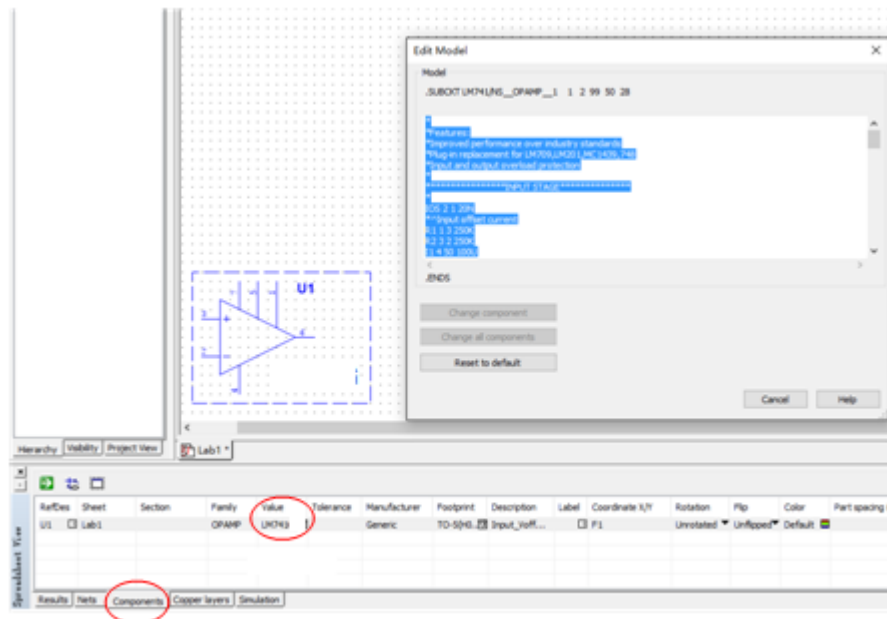


Fig.1 OP07GJ spice model in Multisim.

2.1 DC Open Loop Transfer Characteristic

Simulate the circuit in Fig. 2, sweeping V_{in} from -5 mV to $+5$ mV. Plot V_{out} as a function of V_{in} and determine the open loop gain of the op-amp and input offset voltage. $V_{CC} = 12.0$ V; $V_{EE} = -12.0$ V

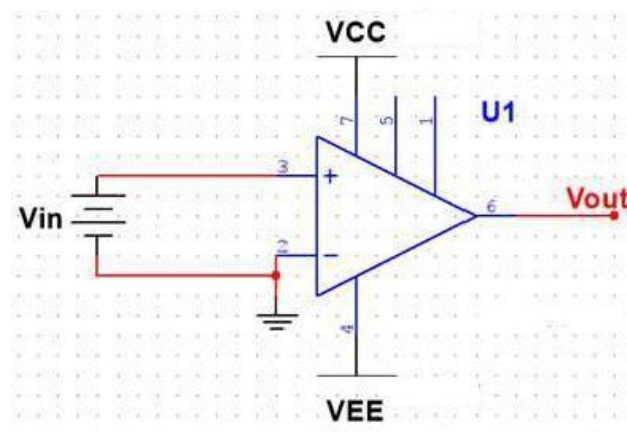


Fig.2 Circuit for Problem 2.1.

Implement the DC sweep function of 2.6 in AD II by Labview programming. This will be used in deriving the dc open loop transfer characteristic in the lab.

2.2 Nulling the Offset Voltage

Null the offset voltage by adding a DC component V_{shift} to the input voltage source to cancel the offset. While this works in simulation, in the lab you will use the offset null terminal pins on the op-amp to null the offset voltage. Check the datasheet for the circuit used to null the offset voltage and draw the circuit connected to the pin out diagram for the OP07G on your pre-lab worksheet. Why do we use a potentiometer to null the offset voltage?

2.3 Slew Rate Measurement in Unity Gain Configuration

Simulate the circuit in Fig. 3 to look at the transient response of V_{out} to the square wave voltage source V_{in} . Set V_{in} to have the following characteristics: an amplitude of ± 2.5 V, a 1 μ s delay, 100ns rise and fall times, and a pulse width and period of 100 μ s and 200 μ s. In your interactive simulation, set the maximum time step to be 100 ns and make sure to simulate one period of the square wave. Plot V_{out} and V_{in} versus time and determine the slew rate (remember: the slew rate has units of $V/\mu s$) from the plot. Is the slew rate different for rising and falling voltages?

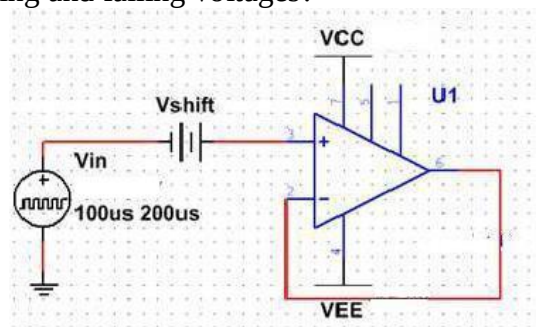


Fig. 3 Circuit for Problem 2.3.

2.4 Gain and Bandwidth in Unity Gain Configuration

Simulate the circuit in Fig. 4, performing an AC analysis from 1 Hz to 1 GHz. Remember to set the magnitude of V_{in} (V1) to 1 V to get the proper value of voltage gain and to plot enough points to get a smooth curve. Plot the magnitude $20\log|V_{out}/V_{in}|$ response, and find the gain (A_0) and 3 dB bandwidth of the amplifier.

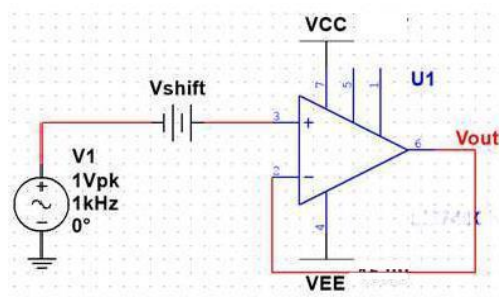


Fig. 4 Circuit for Problem 2.4.

2.5 Gain and Bandwidth in Non-Inverting Amplifier Configuration

Simulate the circuit in Fig. 5 with $R = 100$ k Ω and $R = 10$ k Ω , performing AC analyses from 1 Hz to 1 GHz. Plot the magnitude response of both circuits in dB on the same plot as Problem 2.4, and find the gain and 3 dB bandwidth of both amplifiers. At approximately what frequency and gain do the three curves intersect on the plot? What

does this mean? If you are unsure, compute the product of A_0 and f_{3dB} for the three amplifiers.

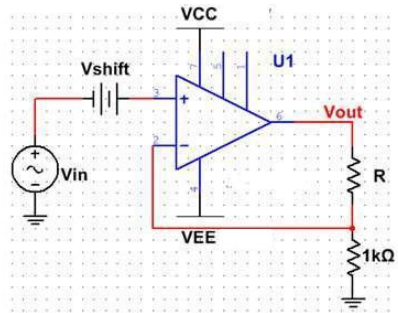


Fig. 5 Circuit for Problem 2.5.

2.6 DC sweep in ADS

Implement the DC sweep function in ADS by Labview programming. **This will be used in deriving the dc open loop transfer characteristic in the lab.**

In this program, functions followed should be implemented:

1. Choose an analog output port to control V_a by DAQ.
2. Calculate V_{in} by the value of V_a .
3. Choose an analog input port to measure V_{out} by DAQ.
4. Use a loop to measure V_{out} vs. V_{in} , a step voltage should be added to V_a in each loop, which implement V_a sweep and you can choose the begin and end voltage.
5. Plot graph of V_{out} vs. V_{in} .

3. Lab

3.1 DC Open Loop Transfer Characteristic

Because the gain of the op-amp is so high, (the output voltage can swing from $-$ rail to $+$ rail when sweeping the input voltage by 10 mV, the smallest voltage step on the Variable Voltage Supplies in ADS) the direct gain measurement is not feasible. We first must attenuate the input voltage before feeding it into the input of the op-amp. Before wiring up the circuit in Fig. 6, Design the resistance of the two resistors in the voltage divider circuit. Measure the practical values using the multimeter and write the resistance values in your Lab Worksheet.

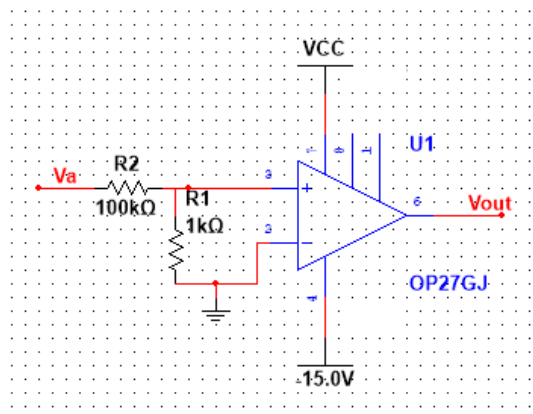


Fig.6 Circuit for Problem 3.1

Since there is no existing DC sweep function in ADS, you need to implement the DC sweep function in ADS by Labview programming. Sweep V_a over a large

voltage range to start out with and then sweep V_a over a smaller range so you can get enough points to be able to accurately calculate the gain A_0 . Plot V_{out} versus V_a based on the discrete points. Determine A_0 and the input offset voltage V_{off} from the plot and write them in your lab worksheet. **Make sure to back-calculate V_{in} from V_a before determining the gain and input offset voltage!**

3.2 Nulling the Offset Voltage

Connect the op-amp in the configuration shown in Fig. 7. Make sure to connect the middle lead (“the wiper”) of the 10 k Ω potentiometer to the -12 V power supply rail. Ground V_{in} and connect V_{out} to the multimeter and measure the output voltage of the op-amp. Adjust the potentiometer until the measured output voltage is nulled as best as you can. Measure the minimum voltage V_{out} change from positive to negative and V_{out} change from negative to positive.

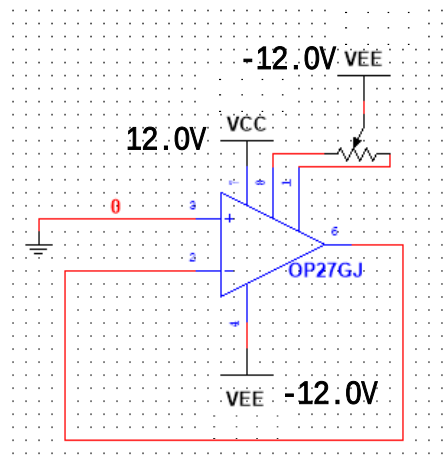


Fig. 7 Circuit for Problem 3.2

3.3 Slew Rate Measurement in Unity Gain Configuration

Keep the op-amp in the configuration as shown in Fig.8. Apply a square wave signal to the input of the op-amp by connecting V_{in} to the function generator.

Considering the output impedance of the uncton generator and the difference between the preset output voltage and the practical output voltage. See the function generator tutorial if you are confused about it. Monitor V_{in} and V_{out} on the oscilloscope on separate channels. Apply a 10 V V_{pp} square wave to V_{in} and determine the slew rate for both the rising and falling edge from the oscilloscope trace. Write the slew rates in your lab worksheet, and attach the oscilloscope trace(s) to the end of your lab worksheet.

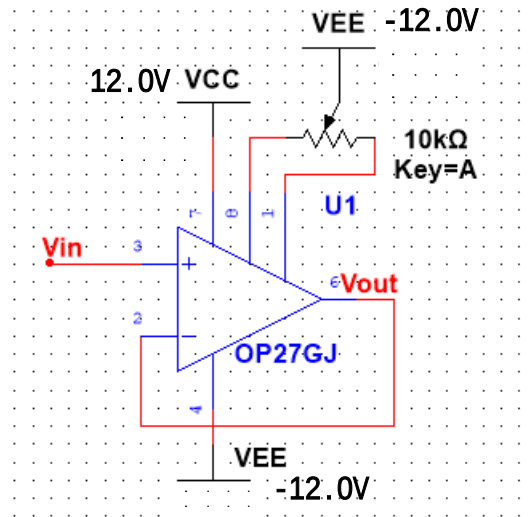


Fig. 8 Circuit for Problem 3.3, 3.4

3.4 Gain and Bandwidth in Unity Gain Configuration

With your op-amp in the same configuration as shown in Fig.8, change the function generator to generate a sine wave signal. Set the amplitude and frequency of the signal high enough so that it is intentionally slewing. This means that your sine wave output needs to have:

$$\frac{dV_{out}}{dt} = 2\pi A \cdot f \geq \text{Slew Rate}$$

Save the oscilloscope trace of the slewing output voltage. Now that you know what a slewing output voltage looks like, reduce the amplitude of the sine wave signal to ensure that it will not produce a slewing output. Measure the magnitude response of the voltage gain of the circuit from 10 Hz to 3 MHz, making sure to plot at least 3 points per frequency decade. If you can measure out to f3dB, write it in your lab worksheet. Also, write the measured gain A0 in your worksheet.

3.5 Gain and Bandwidth in Non-Inverting Amplifier Configuration

Now connect your op-amp in the configuration shown in Fig. 9. For both $R = 10\text{ k}\Omega$ and $R = 100\text{ k}\Omega$, Using the Bode Analyzer in ADS to derive the magnitude response of the voltage gain of the circuit from 10 Hz to 3 MHz. Compare the magnitude response of the voltage gain in dB for the two non-inverting amplifier circuits and the circuit from Problem 3.4 and then give your conclusions.

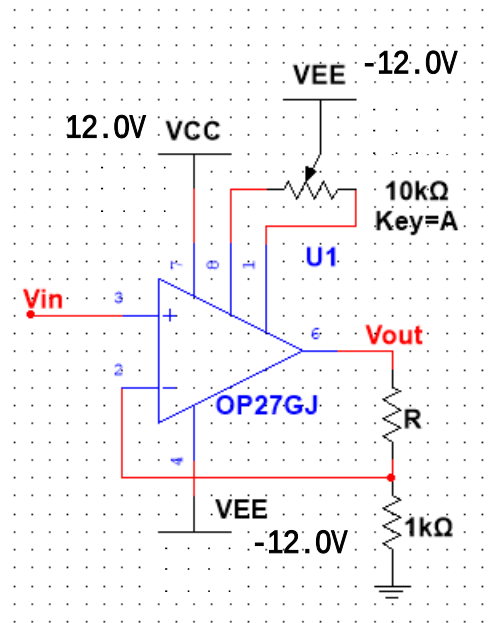


Fig. 9 Circuit for Problem 3.5

Reference

[1] UNIVERSITY OF CALIFORNIA AT BERKELEY, College of Engineering
Department of Electrical Engineering and Computer Sciences, EE105 Lab
Experiments